

D3 — SOFTWARE ARCHITECTURE DOCUMENT

1. Introduction

This document presents the high-level architecture of the SafeSight system. The objective is to define the structural organization of the system and explain how architectural decisions satisfy both functional and non-functional requirements identified in the Software Requirements Specification (D2).

SafeSight is a real-time industrial safety monitoring system that processes live video streams, detects PPE compliance violations and hazardous conditions, and manages incidents through an integrated workflow.

2. Architecture Drivers

2.1 Functional Drivers

The following requirements strongly influence the architecture:

- FR-01: Process live video streams in real time
- FR-02/03: Detect PPE and hazard objects using YOLO11
- FR-04: Generate alerts based on detection thresholds
- FR-05: Automatically create incident records
- FR-06: Validate incidents by Safety Officer
- FR-07/08: Assign corrective actions and update incident status
- FR-09: Generate compliance reports
- FR-10: Enforce role-based access control for all authenticated users
- FR-11: Store evidence snapshots for confirmed incidents
- FR-12: Maintain audit logs for critical actions (validation, status updates, configuration changes)

These requirements collectively drive the need for continuous stream processing, an event-driven workflow, a role-aware security layer, persistent evidence storage, and a comprehensive audit trail.

2.2 Non-Functional Drivers

- NFR-01 Security: Authentication required before accessing any protected system function
 - NFR-02 Authorization: Role-Based Access Control (RBAC) restricts actions by user role
 - NFR-03 Performance: Alerts must be generated within 2 seconds of detection under normal operating conditions
 - NFR-04 Reliability: Automatic camera reconnection and fault tolerance after stream interruption
 - NFR-05 Maintainability: Detection thresholds configurable without modifying application code
 - NFR-06 Auditability: Audit logs maintained for all critical operations (validation, status updates, configuration changes)
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3. Architecture Style

SafeSight adopts a hybrid architecture combining three complementary styles, each chosen to address a specific set of requirements:

- **Layered Architecture** — enforces a clear separation of concerns across the Presentation, Application, Processing, and Data layers. Each layer has a well-defined responsibility, which improves testability and maintainability (NFR-05) and allows each tier to evolve independently without affecting adjacent layers.
- **Client–Server Architecture** — separates the web-based dashboard (Safety Officer, Supervisor) from the backend services. This decoupling supports role-based access enforcement (FR-10, NFR-02) and allows the user interface to be updated or replaced without modifying the detection or incident management logic.
- **Event-Driven Architecture** — governs the real-time detection pipeline. The Detection Engine continuously produces detection events; the Alert Manager and Incident Manager react to these events asynchronously. This pattern is essential to meet the 2-second alert generation target (NFR-03, FR-04) and naturally supports the sequential safety workflow from detection through notification and incident creation.

Together, these three styles ensure that SafeSight remains responsive under continuous video load, maintainable as detection classes or workflows evolve, and structurally aligned with the functional and non-functional requirements established in D2.

4. High-Level Architecture

Layers:

Presentation Layer

- Web dashboard (Safety Officer, Supervisor)

Application Layer

- REST API
- Business logic services

Processing Layer

- YOLO11 Detection Engine
- Alert Evaluation Logic

Data Layer

- Relational DB (Incidents, Users)
 - File Storage (Evidence)
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5. Component Breakdown

Camera Input Service

- Receives RTSP/USB streams
- Sends frames to detection engine

Detection Engine (YOLO11)

- Detects: helmet, vest, gloves, goggles, person, fire, smoke, forklift
- Outputs detection events (confidence + class)

Alert Manager

- Evaluates detection confidence
- Applies threshold logic
- Generates alert events

Incident Manager

- Creates incident records
- Manages incident lifecycle (open → confirmed → resolved / false alarm)
- Stores status updates

Notification Service

- Sends alerts to users (email/push)

User Management Service

- Authentication

- Role-based authorization

Reporting Service

- Generates KPI reports

Storage

- Database → structured data
 - File system → evidence snapshots
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6. Interfaces and Communication

Source	Target	Type
Camera	Detection Engine	Video Stream
Detection Engine	Alert Manager	Event
Alert Manager	Incident Manager	API/Event
Incident Manager	Notification Service	API
UI	API	REST
API	Database	SQL
Reporting Service	Storage	SQL
User Management Service	Database	SQL

Data Flow Overview

The system operates in a sequential event-driven flow:

1. Camera System streams video to the Detection Engine.
2. Detection Engine produces detection events.
3. Alert Manager evaluates events and generates alerts.
4. Incident Manager creates and updates incidents.
5. Notification Service sends alerts to users.
6. Data is stored in the Database and File Storage.

This flow ensures real-time processing and consistent incident lifecycle management.

7. Cross-Cutting Concerns

Security

- Authentication (NFR-01): All users must authenticate before accessing any protected function. The system uses session-based token authentication; unauthenticated requests are rejected with an appropriate error response.
- Authorization (NFR-02): Role-Based Access Control (RBAC) is enforced at the API level. Four roles are defined — Safety Officer, Supervisor, Worker, and System Administrator — each with a distinct permission set. Critical operations such as incident validation (FR-06), corrective action assignment (FR-07), and configuration updates (UC-08) are restricted to their designated roles. Unauthorized access attempts are logged and rejected.

Logging

- The system maintains structured audit logs for all critical operations (NFR-06, FR-12). Logged events include: incident validation decisions (confirm/reject), incident status transitions, corrective action assignments, configuration and threshold updates, and authentication events. Each log entry captures the actor (user ID, role), timestamp, action type, and affected entity. Log records are stored in the relational database and are accessible only to authorized roles (Safety Officer, System Administrator).

Error Handling

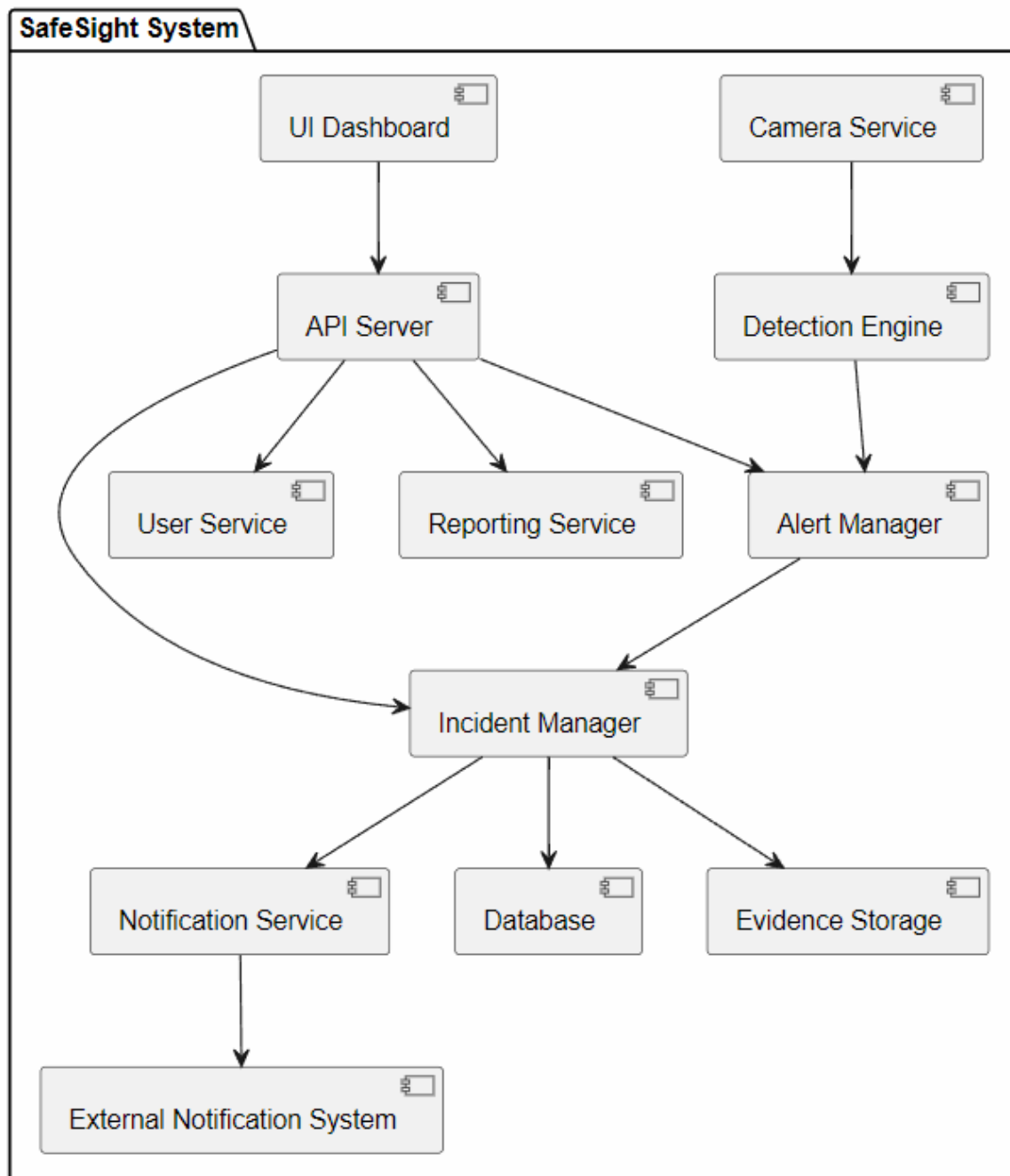
- Camera disconnection (NFR-04): The Camera Input Service monitors stream health continuously. On disconnection, automatic reconnection is attempted at defined intervals. A "camera offline" alert is raised and logged if reconnection fails within the retry window, ensuring Safety Officers are aware of monitoring gaps.
- Notification failure: The Notification Service applies a retry strategy for failed email or push deliveries. Failed delivery attempts are logged with timestamp and recipient for audit purposes.

Detection pipeline failure: If the Detection Engine becomes unavailable, frame processing is suspended and a system health alert is generated. No false alerts are produced during degraded operation.

8. Component Diagram

The component diagram illustrates the high-level structure of SafeSight. It shows the interaction between core components including detection, alert generation, incident management, and external notification services.

The architecture supports real-time processing and modular scalability.



9. Traceability

The following tables map each functional and non-functional requirement from D2 to the responsible architecture component.

Requirement	Description (from D2)	Architecture Component
FR-01	Process live video streams in real time	Camera Input Service

• FR-02	Detect PPE elements (helmet, vest, gloves, goggles)	Detection Engine (YOLO11)
FR-03	Detect hazard objects (fire, smoke, forklift)	Detection Engine (YOLO11)
FR-04	Generate alerts on threshold violation	Alert Manager
FR-05	Automatically create incident records	Incident Manager
FR-06	Validate incidents (confirm/reject) — Safety Officer	Incident Manager
• FR-07	Assign corrective actions — Supervisor	Incident Manager
FR-08	Update incident status until closure	Incident Manager
FR-09	Generate weekly/monthly compliance reports	Reporting Service
FR-10	Enforce role-based access control (RBAC)	User Management Service
FR-11	Store evidence snapshots for confirmed incidents	Storage (File System)
FR-12	Maintain audit logs for critical actions	Logging / Storage (DB)

NFR ID	Quality Attribute	Architecture Component(s)
NFR-01	Security — Authentication	User Management Service
NFR-02	Authorization — RBAC	User Management Service + REST API
NFR-03	Performance — Alert $\leq$ 2 seconds	Detection Engine + Alert Manager
NFR-04	Reliability — Camera auto-reconnect	Camera Input Service
NFR-05	Maintainability — Threshold config without code change	Alert Manager
NFR-06	Auditability — Audit log for critical actions	Incident Manager + Logging + Storage (DB)